ASPRS Guidelines on Quantifying Horizontal Sampling Density of Aerial Lidar Point Cloud Data - EDITION 1, VERSION 1.0 – JANUARY 2025

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1. Introduction and Scope

The density of an unorganized point cloud, specifically one acquired from airborne lidar, is a fundamental characteristic of lidar data. Density is used as a measure to evaluate coverage with the number of points within an area of interest in lidar data collections. It is also used as input into many aspects of data processing, including machine learning processes. The fidelity of information extracted from the dataset has been shown to improve as the point density increases. However, the computational requirements, labor efforts, and costs may also increase.

The use of a single statistic alone is inadequate to describe the density of a lidar dataset. Lidar density is a statistical distribution, not always easily summarized by a single number. Reporting a large point cloud dataset as a single density value ignores the variance inherent in most lidar systems due to airborne collection variables and ground cover. Large variation in point density - more so on the lower end of the distribution - could warrant a rejection of the data if it violates the contractual requirements. A common understanding of lidar data density is therefore crucial in making decisions during procurement regarding collection criteria, at the post-collection QA/QC stage, and for the end user. This understanding must include a reliable method for quantifying and reporting point density.

The method used to measure density has been shown to have an impact on the value of derived products. Therefore, an understanding of various common methods used to measure density is also valuable. Dropping below a minimum point density threshold (regional or localized) would likely require supplemental collected data to meet specifications. Density testing is required to determine the acceptance of data. Unfortunately, the most commonly used horizontal sampling methods can often produce widely varying results when compared to one another due to inherent differences and limitations in their measuring process. An accurate and consistent horizontal density measurement methodology is needed to produce point cloud density statistics and pass/fail thresholds for aerial lidar.

This report provides a common understanding of the definitions of lidar data density, methods to quantify the density, the tradeoffs, and recommendations. The goals of this paper are to:

- Clarify various definitions of density and related terms
- Document various methods for quantifying, representing, and reporting density
- Document tradeoffs among the methods of quantifying, representing, and reporting density
- Provide recommendations on the ASPRS preferred method of density measuring and reporting

Not intended to be addressed in the first version of this document:

- Elevation based point density (building walls, trees, etc.) by way of a cubic voxel. Some 2D density results could be inflated in localized areas if used to measure airborne oblique lidar in regions with buildings. This paper focuses on the horizontal sampling of point cloud data.
- The limiting factor on density is based on the laser's illuminated footprint diameter. One can argue that significantly overlapping laser footprints are not contributing to the point density by virtue of redundant measurements. This paper is not currently focused on the need to prefilter data based on nonoverlapping measurements.

2. Supporting Terms and Definitions

- *active remote sensor* – a system designed to detect the target surface through emission of electro-magnetic energy and the subsequent capture of this energy reflected from the surface.

- *aliasing* - the distortion or artifact in a dataset that occurs due to the sampling pattern of a LiDAR sensor, particularly when the point spacing is not uniform or when the sampling rate does not adequately capture the details of the terrain or objects being scanned.

- *passive remote sensor* – a system designed to detect the target surface through capture of electro-magnetic energy from sunlight reflected off the surface or of energy emitted directly from the target surface.

- *Point spacing uniformity* - The uniformity of the spatial distribution of the points within a point cloud. Point clouds that are not isotropic, or may change with the direction of measurement, do not provide a consistent representation of the scanned surface with consistent point spacing in all directions. Uniform point spacing is crucial for accurate surface modeling, reducing artifacts, aliasing, and ensuring that all features are sampled with similar density across the survey area.

- *raster (a.k.a. image, grid, or array)* – a two-dimensional grid of pixels in a coordinate reference system where each pixel may store discrete value(s)

- *spatial resolution* – At its most basic level, spatial resolution of the point cloud expresses the size of a surface feature that can be resolved from within the point cloud. Spatial resolution describes the fidelity or fineness of detail of the point cloud. Spatial resolution is defined by two components: point proximity and point detection region.

    - *Spatial Resolution, Point Proximity* – The spatial relationship of a point to neighboring points. This metric may be considered analogous to point density and point spacing. The closer the proximity of a point to neighboring points, the greater the point density and the lower the point spacing.

    - *Spatial Resolution, Point Detection Region* – This component of spatial resolution pertains to the area of the surface represented by the point. For active remote sensing technology, point detection region is defined by either the spread of backscattered energy reflecting off the surface back to the sensor detection system, or the instantaneous field of view of the sensor detector element. For active remote sensing technology, the larger the point detection region, the more ambiguous the position of the point in 3D space. Conversely, the smaller the point detection region, the more precise the position of the point in 3D space.

      For passive remote sensing (EO imaging), point detection region is defined by the dimension of the projected detector element (pixel Ground Sample Distance) and the quality and quantity of pixels used in the correlation process.

- *tenderloin/overage* - Tenderloin points are the most geometrically reliable points of a swath that end at another swath's tenderloin or a project boundary. Inversely, overage points are less

geometrically reliable points at the edges of swaths that overlap other swath's tenderloin. For projects with multiple swaths, tenderloins may overlap neighboring swaths tenderloins in order to achieve an aggregate density utilizing multiple swaths.

- *vertical point density* – The number of points from a point cloud per unit of area. The vertical density metric defines the spatial relationship between points in three dimensions (horizontally and vertically). Proper use of the term "vertical point density" or "vertical density" requires the qualifying descriptors "design" or "tested", as well as the "point type".
  Alternate definition: The vertical density metric defines the spatial relationship between points in the vertical dimension.

- *Voronoi diagram* – A tessellation formed by lines which bisect and are perpendicular to each triangle edge of a Triangulated Irregular Network (TIN) model.

3. What is Point Density?

3.1 Point Density – The number of points from a point cloud per unit of area. The density metric defines the spatial relationship between points in two dimensions (horizontally). Proper use of the term "point density" or "density" requires the qualifying descriptors "design" or "tested", as well as the "point type". Point Density is inversely related to the square of Point Spacing as follows:

$$Point\ Density = \frac{1}{(Point\ Spacing)^2}$$

Typically, the phrase "number of points per square meter" is used.

3.2 Point Spacing – The point spacing metric enables us to answer a fundamental query in the analysis of (unstructured) point data sets: What are the nearest neighbors of a (2D or 3D) given point, and how far are they?
For unstructured point data, the distance between points will vary. The variation is both spatial and directional. Therefore, to bring some structure to this important metric, we define Point Spacing as the inverse of the square root of Point Density:

$$Point\ Spacing = \frac{1}{\sqrt{Point\ Density}}$$

This definition is isotropic (i.e., does not change with the direction of measurement), while allowing for spatial variation in the metric across the entire dataset.

3.3 Point Type – The qualifying term to define which points are used in the density calculation (a.k.a. "indicator points"). Point type terms pertinent to this paper are as follows:

- All return points - Points of any return type produced by lidar systems that capture return information.
- First return points – Return values "one of one" or "one of many" produced by lidar systems that capture return information.
- Last return points - Return values "one of one" or "last of many" produced by lidar systems that capture return information.

- Pre-conditioned points – Usually pertaining to points from a lidar system that does not record return information (e.g., Geiger mode lidar). Points that have been selected by a preprocessing step to be used in a computation such as density calculation. In this paper, one of the methods uses pre-conditioned points by selecting only locally planar surfaces in three-dimensional space.
- Ground points – Points classified to ground excluding any non-ground features. This classification is typically generated from an automated filter followed by manual check and corrective edits to result in an accurate depiction of the ground.

3.4 Design Point Density – The anticipated density metric resulting from the data capture system. For lidar point cloud data, the design point density is a function of sensor speed over ground, pulse repetition frequency, scan angle range, scan frequency, and distance to target. These things can be planned prior to data capture and can therefore be used to define the design point density. Requires use of the "point type" term.

3.5 Tested Point Density – The actual density metric product from a spatial analysis routine. The simplest density metric can be calculated by dividing the number or points of a specific point type by the two-dimensional area of the tested region. A more complex density metric can be derived from the use of Voronoi polygons to determine the area represented by each point. Any reported tested point density needs to specify the point type, statistical representation, and spatial extent. The statistical representation should include the statistical method, the sampling frequency, and sampling dimension.

For example:

- the mean ground point density in a tile of data is 5 ground points per 4 sq meters
- the first return point density has a mode of 2 first returns per 1 sq meter using Voronoi diagrams within a 100 sq meter pixel raster

It needs to be understood that these are different questions:

- What is the density of a point cloud dataset?
- How does the density of a point cloud dataset vary across the project area?
- Does a dataset meet a minimum requirement point density? If the answer is no - how and where does point density fail?
- Can a point cloud dataset be used for an intended application? What minimum density is required for that use?

Lastly, point density testing should occur after spatial distribution verification (e.g., acceptable along track vs cross track "squareness") has been verified and data void~~s~~ checks have been completed and patched if necessary.

4. Requirements for Lidar Point Density Measurement and Reporting

- Sensor and scanner type agnostic – recognizing that data from lidar systems that record return information can and should be preprocessed differently than those than those that lack return attribution to generate indicator points to be used as input into the density calculation.
- Must be able to accommodate datasets that have been flown with multiple channels/sensors or passes to achieve the intended minimum point density.
- Results need to be easy to interpret and reporting be meaningful with respect to the entire project.
- Quantitative (pass/fail) and qualitative (visual) results

- Pass/fail results cannot be biased (inflated) by overlap designed only for coverage (as opposed to contributing to the required density), crosslines, or collection block overlap.
- For aerial non-bathymetric lidar, water areas should be excluded from the result.
- Need project-wide measurement and reporting that can identify density problems/failures anywhere rather than relying on representative samples that can easily miss localized issues.
- A method to be used during post-collection data check, pre-delivery QC, and by end users for density analysis.
- Recommended options:
  - Ability to select point classifications for density testing (e.g., ground density)
  - Ability to exclude water bodies
  - Ability to benefit from high-performance computing (parallel processing, intelligent RAM usage, etc.)

5. Methods for Measuring Density

Density measurements depend on the methods used. The following is a summary of the density measuring and reporting methodologies. Example graphics for each method are taken from the same dataset with a design density of 8 points per square meter (ppsm).

5.1 Representative Sample – The Representative Sample method uses the process of manually or automatedly defining one or more relatively small areas of interest within a project area. Indicator points (first, last, or pre-conditioned for measurement) are counted and that total is divided into the measured polygon's area.

- Methodology
  1. Point cloud data is loaded into a view/map window.
  2. A user-defined polygon is drawn (typically a localized area) (Figure 1).
  3. The area of the polygon is calculated.
  4. The tool counts the number of indicator points (first, last, or pre-conditioned for measurement) inside the polygon.
  5. This number of counted points is divided into the polygon area and the density value is reported, typically in points per square meter (Figure 2).

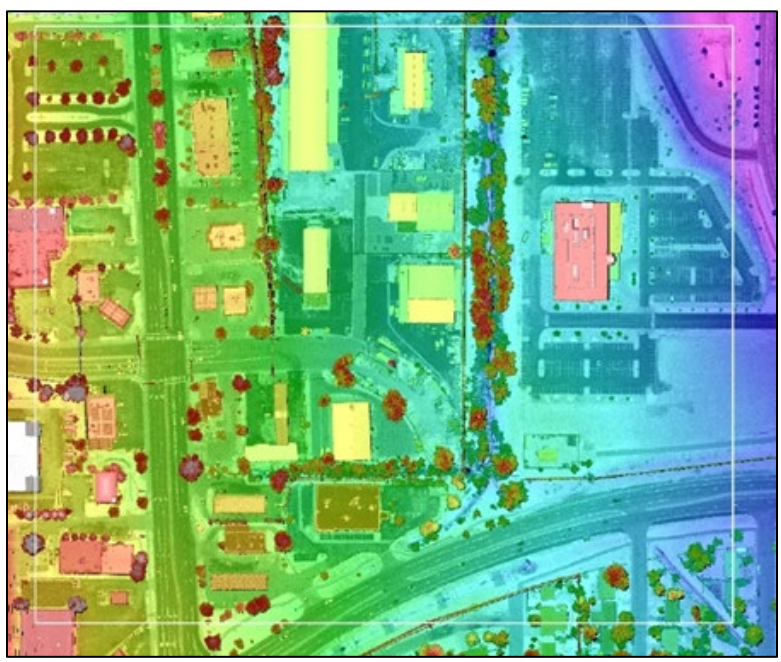

Figure 1: An analysis box drawn around a lidar dataset (courtesy Matt Bethel).

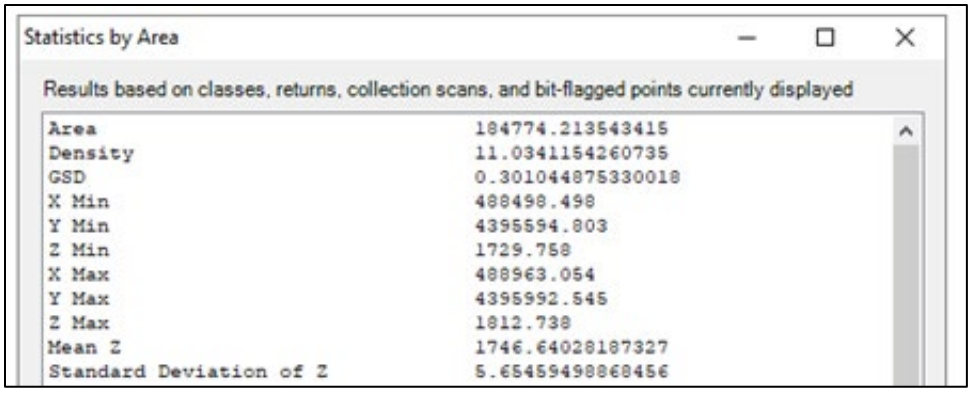

Figure 2: Statistics by Area result of the analysis box above (courtesy Matt Bethel).

- Advantages
  - Fast to calculate with well-known method
  - Works well for localized areas of interest
  - Performs consistently with all lidar scanner types and all acquisition approaches (e.g., >50% overlap or multiple channels/sensors)

- Disadvantages
  - Can be biased (i.e., the results inflated) by many factors such as overlap, cross lines, patches, location in scan, etc.
  - Very localized - not representative of average swath density or overall project density
  - Cannot effectively or automatically find problem areas that could be considered failures/ specification violations
  - Limited use for reporting
  - Not usable for project-wide pass/fail assessment

5.2 Per Swath – The indicator point count (first, last, or pre-conditioned for measurement) is divided into the area of the swath. This is repeated for all swaths.

- Methodology
  1. Lidar swaths are used as inputs.
  2. Each swath's boundary is generated and stored in a polygon format.
  3. The area of each polygon is calculated.
  4. The tool counts the number of indicator points (first, last, or pre-conditioned for measurement) inside each polygon.
  5. This point count is divided into the area and the density product is attributed, typically in points per square meter, for each swath.
  6. Swath densities are reported in a tabular manner for all tested flight lines (Figure 3).

- Advantages
  - Ideal to compare against planned swath density
  - Relatively easy to compute
  - Batchable – one thread per flightline
  - Useful for reporting (based on what is calculated, not what is ideal for overall density)
  - Is not biased (inflated) by overlap
  - Very straightforward

- Disadvantages
  - Does not adequately account for localized density variations such as changes in aircraft speed or sudden variations in pitch.
  - Needs interpretation if flying >50% overlap or multiple passes to achieve planned density
  - Results from lidar systems with inconsistent scanner swath densities can adversely affect the reported density results. Edge clipping may need to be used. Examples of varying density per swath based on the amount of swath used/clipped for density calculations are as follows:

	Points per square meter			
Line number	100% of swath	95% of swath	75% of swath	50% of swath
1	14.76	13.02	11.72	11.40
2	13.85	12.33	11.10	10.77
3	14.14	12.55	11.33	11.06
4	12.96	11.83	10.53	10.30

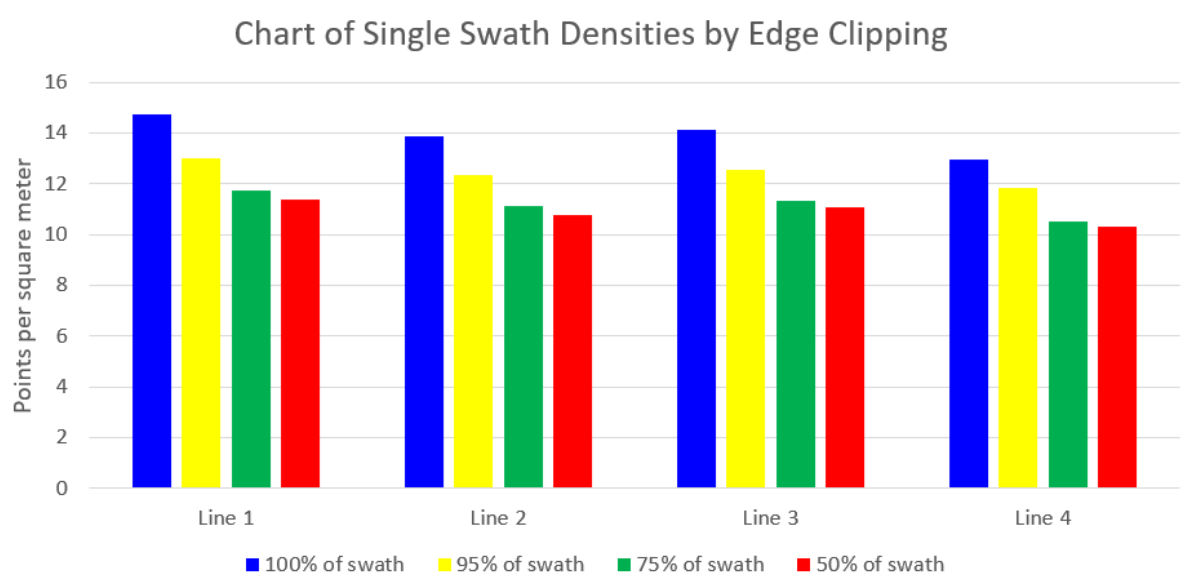

Figure 3: Table and chart of point density by swath (courtesy Matt Bethel).

5.3 Aggregate - The total number of indicator points (first, last, or pre-conditioned for measurement) within a given boundary are divided into the area of that boundary.

- Methodology
  1. A boundary file is selected (project, delivery block, etc.).
  2. The area of the polygon(s) is calculated.
  3. The tool counts the number of indicator points (first, last, or pre-conditioned for measurement) inside the polygon(s).

  4. This point count is divided into the polygon area(s) and the density product is reported, typically in points per square meter (Figure 4).

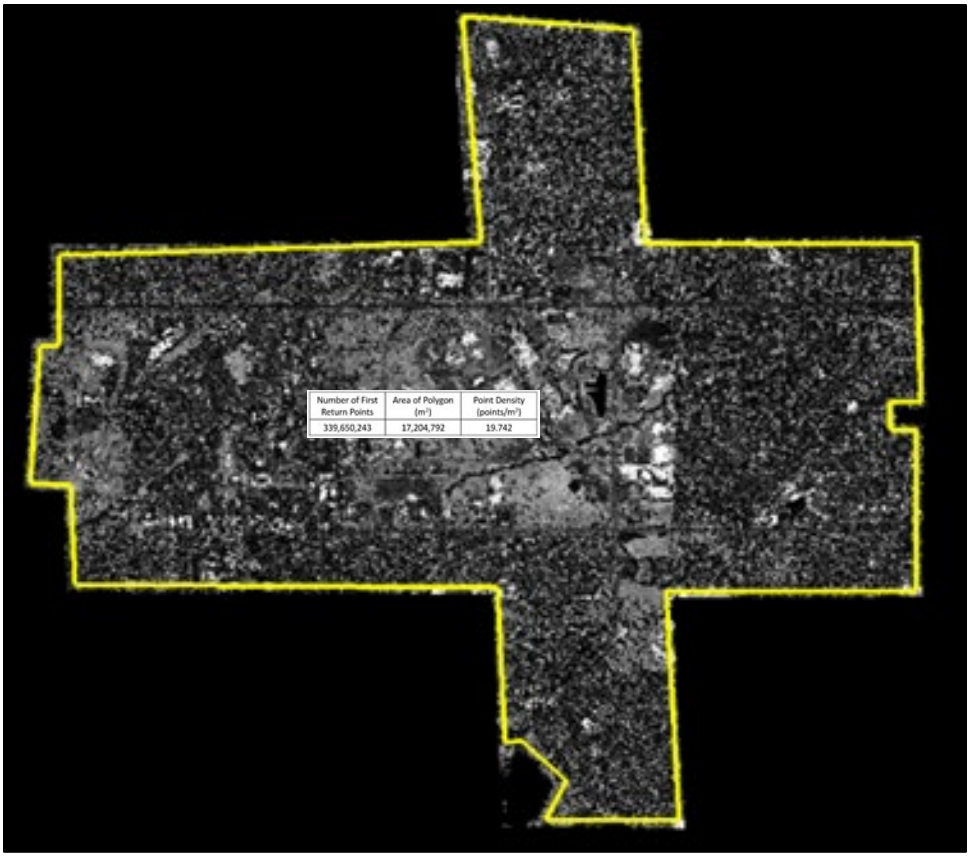

Figure 4: Results of aggregate density by project area (courtesy Matt Bethel).

- Advantages
  - Considers all collected data (selected indicator points)
  - Straightforward approach (the number of indicator points / area of project boundary)
  - Performs consistently with all lidar scanner types and all acquisition approaches (e.g., >50% overlap or multiple channels/sensors)
- Disadvantages
  - A single density value is reported rather than statistics of the distribution of density
  - Swath edge densities, crosslines, overlap, collection block overlap, and patches will greatly inflate the single reported density value
  - Tabular reporting only will not identify localized density failures. A thematic raster is needed for locating potential density issues

5.4 Point in Cell Raster/Tile Count - A raster or tile scheme is used to analyze indicator points. Each cell or tile is attributed with the number of points. These cells or tiles can then be thematically rendered showing point count values. Statistics can be generated from these values.

- Methodology
  1. A tile scheme or raster is generated to cover the point cloud data being measured.
  2. Each tile or raster cell is populated with the number of indicator points (first, last, or pre-conditioned for measurement) inside each tile or cell.
  3. Statistics are generated and reported based on these counts (Figure 5).
  4. A color gradient thematic raster is generated for visual review (Figure 6 and Figure 7).

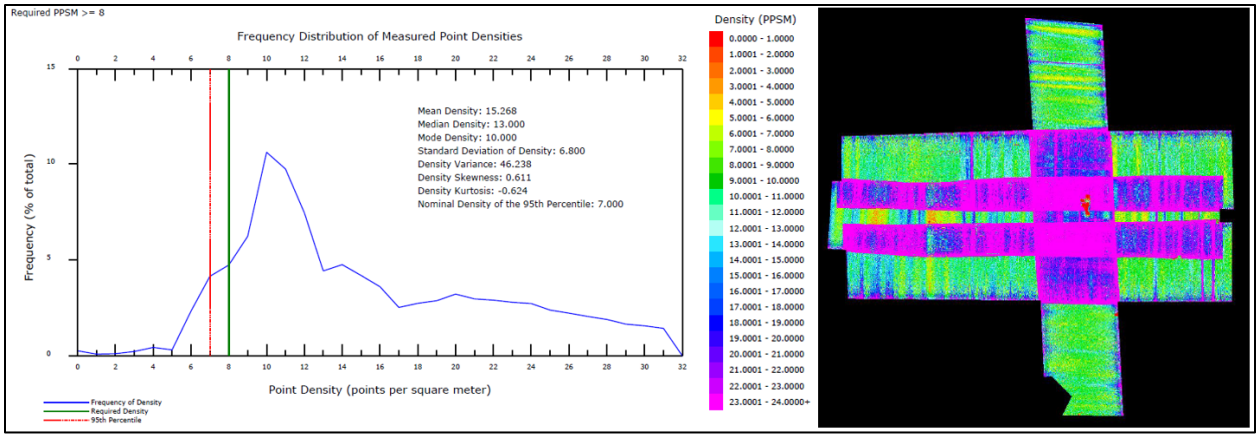

Figure 5: Chart and thematic raster showing point in cell density metrics (courtesy Matt Bethel).

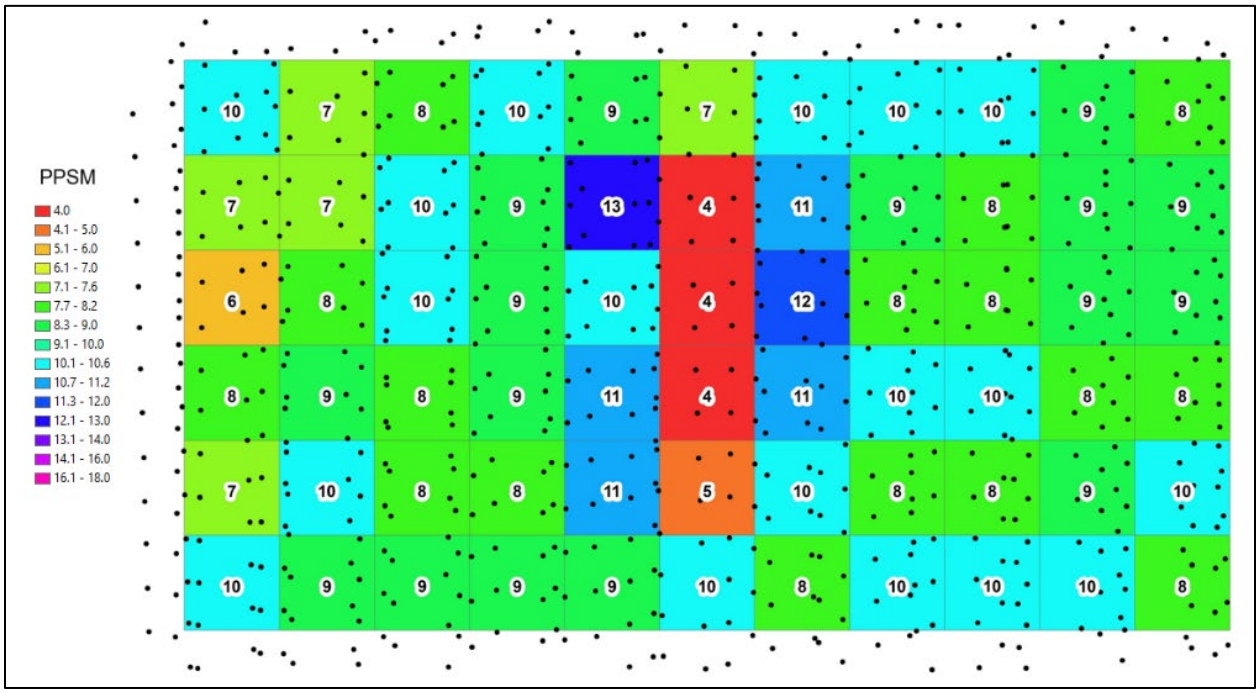

Figure 6: A zoom-in of a thematic rendering of a point in cell raster (8 ppsm lidar / cell size = 1m x 1m) with points overlaid (courtesy Matt Bethel).

Note the cells with counts of 4 have neighboring cells of much higher values because points fall just outside of the 'value=4' cells. This is not an accurate measurement of the point density in this area. This issue can occur throughout the point in cell product skewing the density results. Using varying cell sizes while scaling up the minimum required number of points yields the same result. The other problem with the point in cell method is aliasing. Aliasing in this context is the distortion or artifact that results when measurements of evenly spaced samples are used to create a raster product from randomly spaced points.

While this phenomenon occurs very commonly with naturally, randomly spaced points, it is easiest to see in a point in pixel thematic density raster that is generated from synthetically created point cloud data. If we create a point cloud that has a spacing of exactly 2 ppsm, we expect that the point in cell raster would show us 100% passing results. This is not the case because the point in cell process is flawed as discussed

above. In Figure 7 we see many cells with 1 ppsm (orange), some cells with 2 ppsm (green) and fewer cells with 4 ppsm (blue). With the gridded points overlaid we can see what causes the inconsistent thematic color pattern. Many of the cells have ALMOST two points, but the neighboring pair falls just barely outside of the cell resulting in a count of 1. Likewise, some quadruple sets of points can eventually fit into single cell which provides no "extra credit" for their diagonally neighboring cells of just 1 point. This phenomenon results in an aliasing pattern of cells measuring point counts per square meter. Of course, this aliasing occurs with real Lidar data as well, just in less consistent patterns.

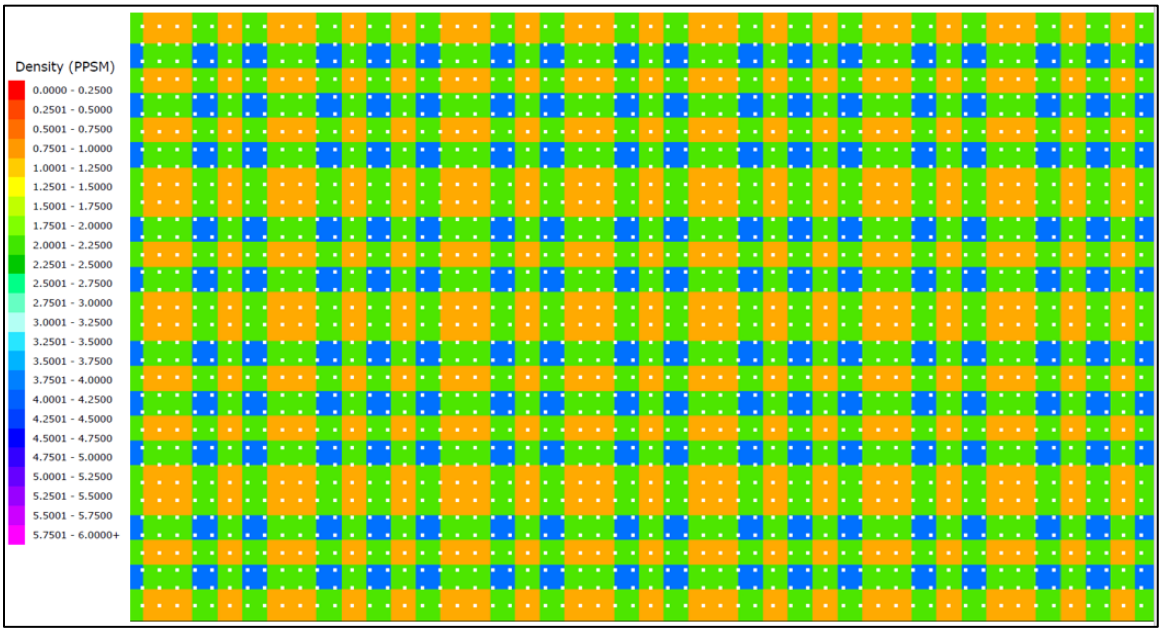

Figure 7: A zoom-in of a thematic rendering of a point in cell raster (synthesized 2 ppsm / cell size = 1m x 1m) with points overlaid (courtesy Matt Bethel).

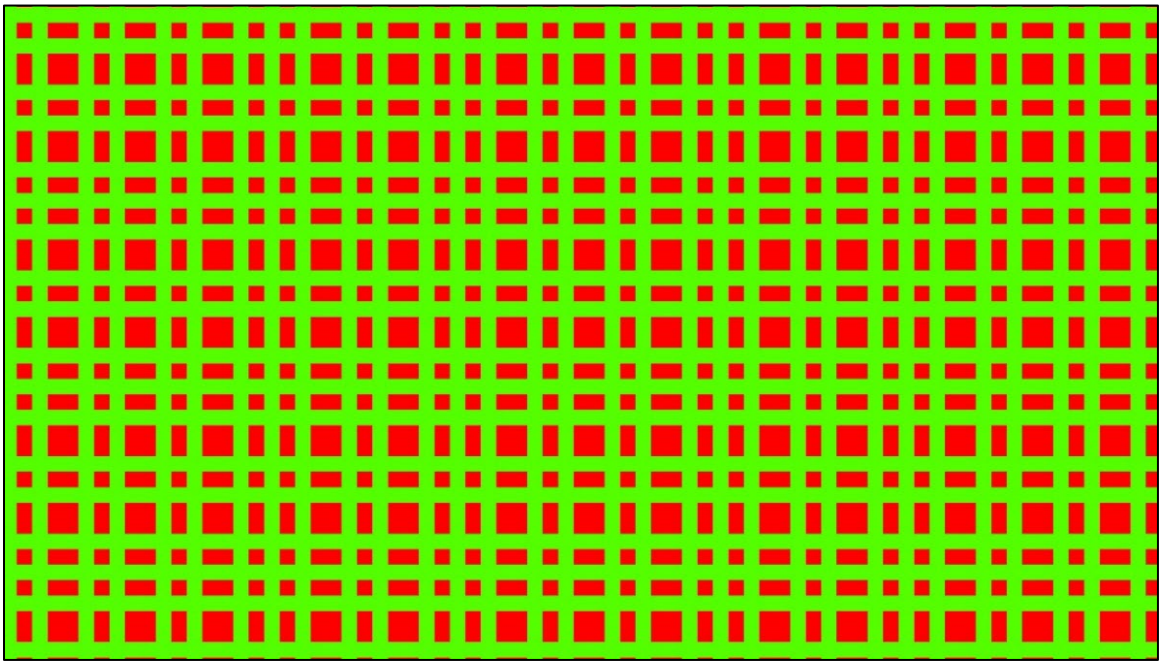

Figure 8: The same area as figure 7 but thematically colored as pass (green) and fail (red) for a minimum of 2ppsm (courtesy Matt Bethel). This is known as a binary raster and will be discussed more in section 5.6.

- Advantages
  - Straightforward approach
  - Widespread adoption
  - Uses a well-known method to calculate and report upon
  - Raster can be thematically rendered for user friendly review
  - Performs consistently with all lidar scanner types and all acquisition approaches (e.g., >50% overlap or multiple channels/sensors)
- Disadvantages
  - Different user-defined processing cells or tile sizes significantly change the results
  - Points at or near tile/cell edges negatively skew the results due to the limited point in cell counts.
  - Inherent with aliasing problems that invalidate the results
  - These disadvantages are most prominent at fringe density cases which coincide with the most cost-efficient way to collect lidar data.

5.5 Compute Statistics for the Sweet Spot of the Swath – This method is used to estimate lidar point density based on computing statistics derived from user-designed cells along the central portion of the swath or tile. This method uses the point in cell raster/tile count method but is isolated to the clipped center portion of swaths.

- Methodology
  1. To evaluate density based on swath by swath, for each swath perform the following:
     A. Design grid of cells within the central 20% of the swath width with *CellX x CellY* size.
     B. Compute point counts (*n*) in each cell, see Figures 9 and 10.
     C. Compute point density for each cell using *Cell Density = n/cell size = n/(CellX x CellY).*
     D. Assign that point's density (*Cell Density)* to that cell.
     E. Generate a raster image of the points density with pixel size is the cell size *CellX x CellY*, here *CellX* is always equal to *CellY.*
     F. Compute the following statistics for each swath:
        i. Mean of *Cell Density* from all cells in the swath, will be referred to as "*Mean of Swath Density"*
        ii. Median of *Cell Density* from all cells in the swath, will be referred to as "*Median of Swath Density"*
        iii. StDEV of *Cell Density* from all cells in the swath, will be referred to as "*StDEV of Swath Density"*
  2. To evaluate project-wide statistics that represent project overall point density, compute the following using results from all the swaths in the project as determined in step (A.) above:
     i. Mean of Swath Density values from all swaths, will be referred to as "*Mean of Project Density*"
     ii. Median of Cell Density from all cells, will be referred to as "*Median of Project Density*"
     iii. StDEV of Cell Density from all cells, will be referred to as "*StDEV of Project Density*"
  - Project-wide statistics are used to represent the overall point density for the project, it will be used for the pass/fail decision according to the contract specified point density.
  - In all the above steps, make sure to mask/ignore outlier cells that are influenced by extraneous swath coverage.

- For project data that is based on tiled point clouds (without access to the swaths data), the same approach can be applied by generating evaluation cells along every vertical column of the tiles or at an interval of columns.

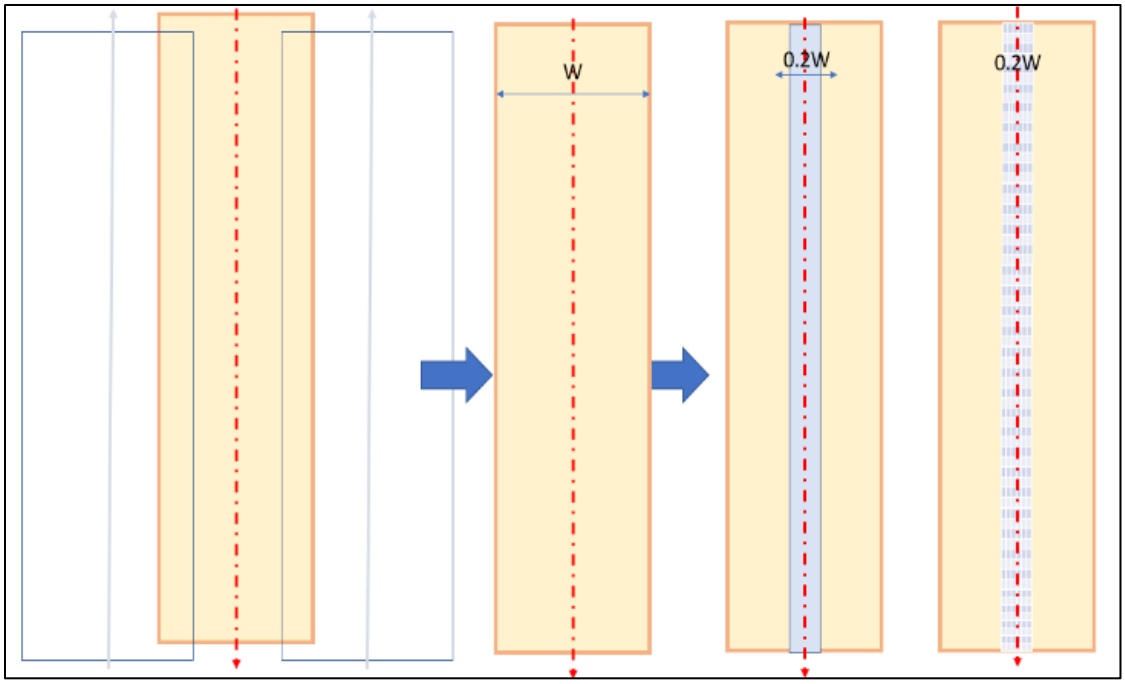

Figure 9: Swath's layout and evaluation cells design (courtesy Qassim Abdullah, Ph.D., CP, PLS)

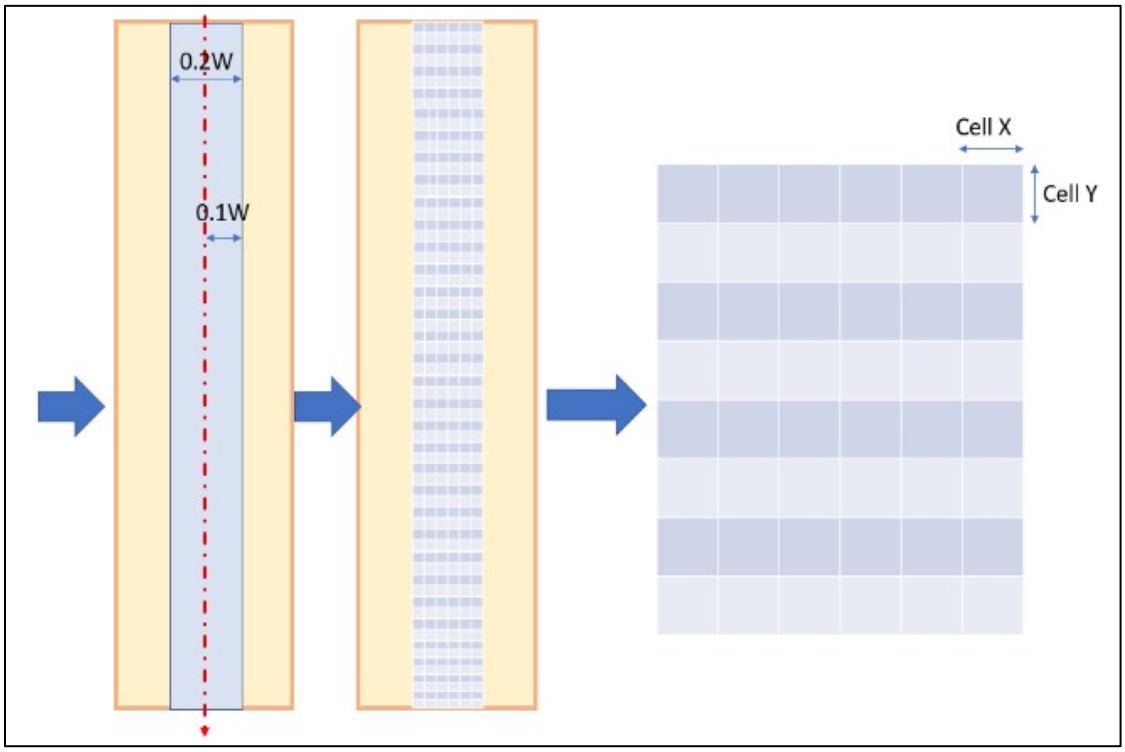

Figure 10: Evaluation cells design (courtesy Qassim Abdullah, Ph.D., CP, PLS)

- Advantages
  - Useful to compare against designed swath density
  - Relatively easy to compute
  - Reasonably batchable – one process per flightline
  - Is not biased (inflated) by overlap nor densification at the edges of some scanners' swaths
- Disadvantages
  - Limited to swath-by-swath analysis, not project-wide nor aggregate point density representation.
  - Needs interpretation if flying >50% overlap or multiple passes to achieve designed density.
  - Focuses on only the central portion of a swath and may hide lower densities found elsewhere.
  - Since it is based on the point in pixel method, it suffers from most of the same flaws:
    - Different user-defined processing cell sizes change the results
    - Inherent with aliasing problems that invalidate the results
    - This group of disadvantages are most prominent at fringe density cases which coincide with the most cost-efficient way to collect lidar data.

5.6 Binary Raster – This testing method is foundationally the same as the point in cell method, but the results are distilled down to only pass and fail values which are colorized and statistically summarized. A tile scheme or raster is used to analyze indicators. Each cell/tile is assigned a pass or fail value based on the point count being greater than or equal to or less than the required point density. See Figure 8.

- Methodology
  1. A tile scheme or raster is generated to cover the point cloud data being measured.
  2. Each tile or raster cell is populated with the number of indicator points (first, last, or pre-conditioned for measurement) inside each tile or cell.
  3. A testing threshold is used, and each tile is assigned a pass or fail value based on the point count being greater than or equal to or less than the required point density.
  4. Percent pass/fail are reported.
  5. A two-color thematic raster is generated for visual review (Figure 11 and Figure 12).

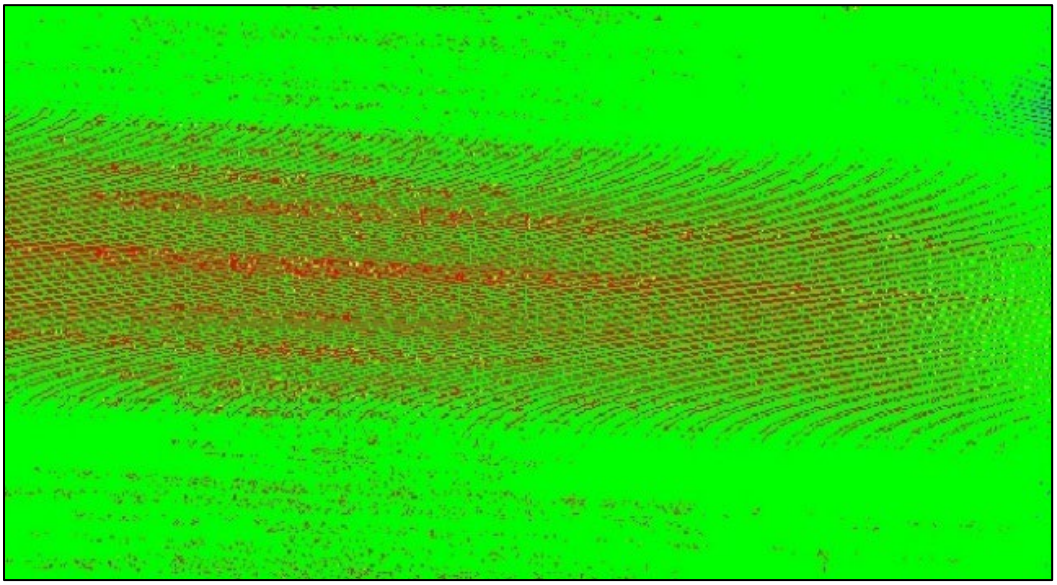

Figure 11: Binary raster (pass/fail) with aliasing (courtesy Matt Bethel).

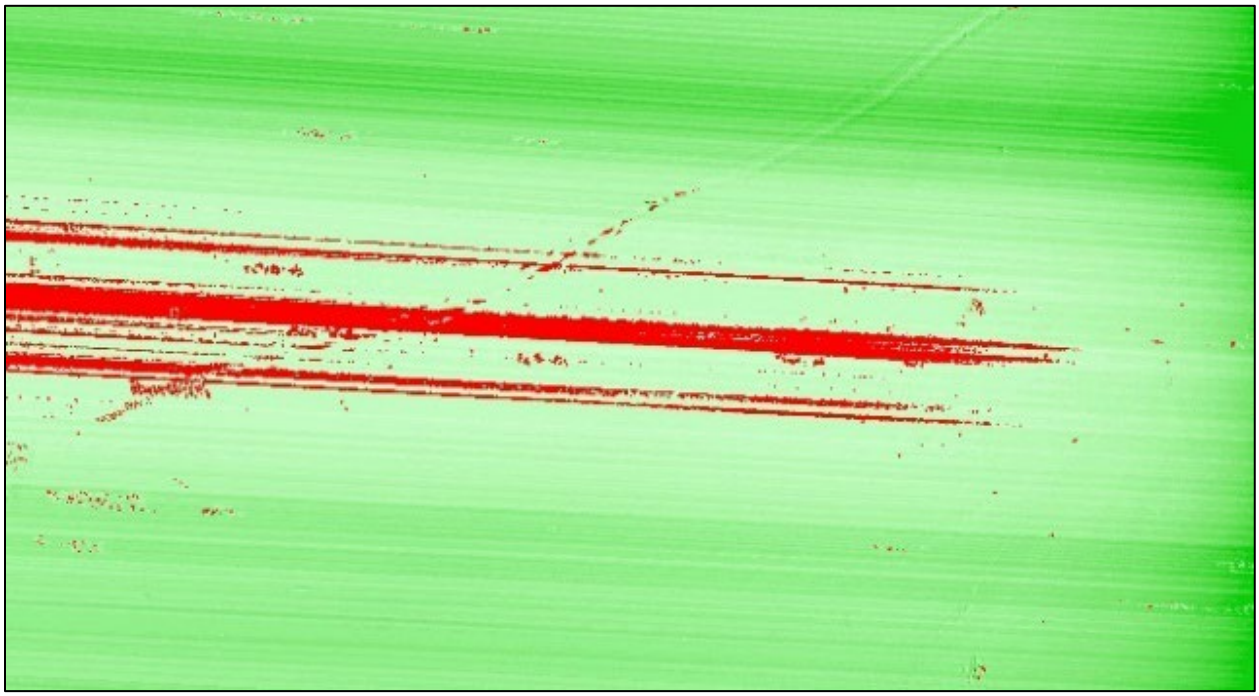

Figure 12: Voronoi pass/fail raster lacking aliasing (courtesy Matt Bethel).

Just like the point in cell method, the binary raster derivative product can result in varying results dependent on the cell size used to count the minimum required number of points, as shown below.

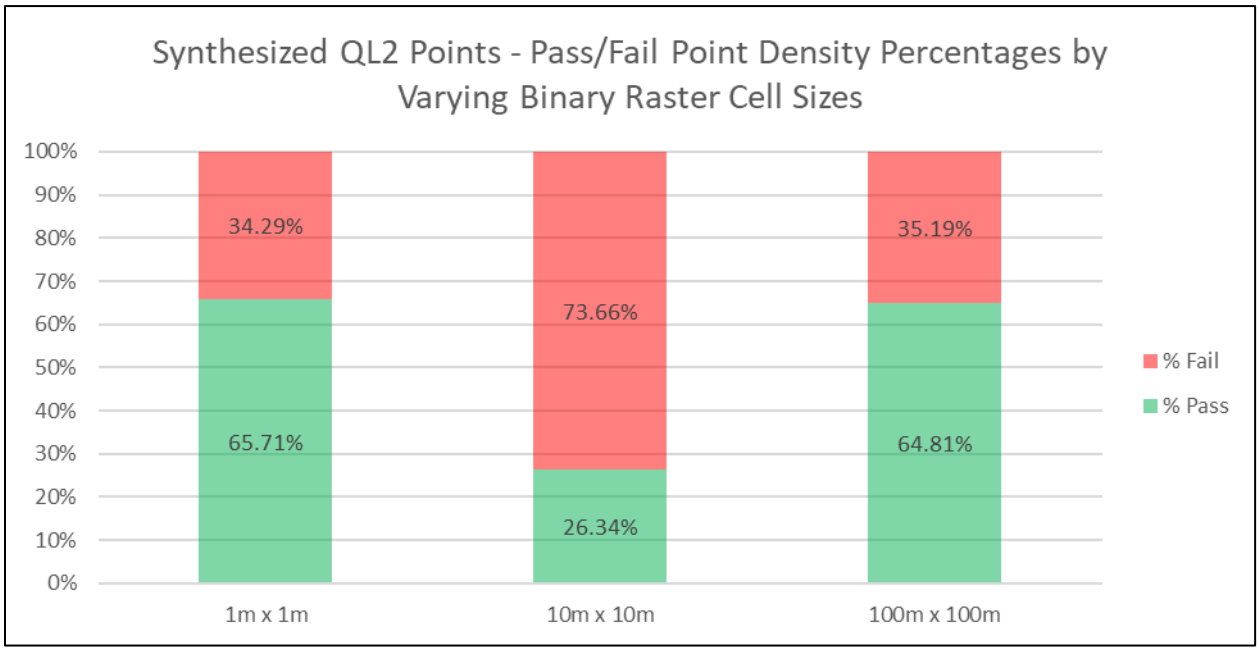

Figure 13: Pass fail results of synthesized 2 ppsm data based on varying test cell sizes (courtesy Matt Bethel).

Binary raster results tested for a minimum of 2 ppsm using actual lidar data (not synthesized). This data passes all non-point in cell tests.

First Return (Pulse) Density

Density computed using a cell that is 1.41w by 1.41h units (2.00 square units)

First return density image is 52099 rows by 2777 columns: test_batch_first_return_density.jpg

Minimum first return density (returns per square unit)	0.50
Maximum first return density (returns per square unit)	444.50
Average first return density (returns per square unit)	2.62
Standard deviation of first return density (returns per square unit)	2.34

Color	Description	Percentage of Area with Data
	Cells with no points (could be outside coverage area)	NA
	Density less than minimum specification (less than 2.00 points per square unit)	24.81
	Density within specification (2.00 to 10.00 points per square unit)	74.02
	Density exceeds specification (more than 10.00 points per square unit)	1.17

First Return (Pulse) Density

Density computed using a cell that is 2.24w by 2.24h units (5.00 square units)

First return density image is 32951 rows by 1757 columns: test_batch_first_return_density.jpg

Minimum first return density (returns per square unit)	0.20
Maximum first return density (returns per square unit)	310.40
Average first return density (returns per square unit)	2.58
Standard deviation of first return density (returns per square unit)	2.13

Color	Description	Percentage of Area with Data
	Cells with no points (could be outside coverage area)	NA
	Density less than minimum specification (less than 2.00 points per square unit)	41.52
	Density within specification (2.00 to 10.00 points per square unit)	57.32
	Density exceeds specification (more than 10.00 points per square unit)	1.16

First Return (Pulse) Density

Density computed using a cell that is 10.00w by 10.00h units (100.00 square units)

First return density image is 7370 rows by 395 columns: test_batch_first_return_density.jpg

Minimum first return density (returns per square unit)	0.01
Maximum first return density (returns per square unit)	72.64
Average first return density (returns per square unit)	2.56
Standard deviation of first return density (returns per square unit)	1.72

Color	Description	Percentage of Area with Data
	Cells with no points (could be outside coverage area)	NA
	Density less than minimum specification (less than 2.00 points per square unit)	48.17
	Density within specification (2.00 to 10.00 points per square unit)	50.94
	Density exceeds specification (more than 10.00 points per square unit)	0.89

First Return (Pulse) Density

Density computed using a cell that is 31.62w by 31.62h units (1000.00 square units)

First return density image is 2331 rows by 126 columns: test_batch_first_return_density.jpg

Minimum first return density (returns per square unit)	0.00
Maximum first return density (returns per square unit)	32.76
Average first return density (returns per square unit)	2.54
Standard deviation of first return density (returns per square unit)	1.40

Color	Description	Percentage of Area with Data
	Cells with no points (could be outside coverage area)	NA
	Density less than minimum specification (less than 2.00 points per square unit)	46.03
	Density within specification (2.00 to 10.00 points per square unit)	53.73
	Density exceeds specification (more than 10.00 points per square unit)	0.24

First Return (Pulse) Density

Density computed using a cell that is 100.00w by 100.00h units (10000.00 square units)

First return density image is 738 rows by 42 columns: test_batch_first_return_density.jpg

Minimum first return density (returns per square unit)	0.00
Maximum first return density (returns per square unit)	9.76
Average first return density (returns per square unit)	2.45
Standard deviation of first return density (returns per square unit)	1.07

Color	Description	Percentage of Area with Data
	Cells with no points (could be outside coverage area)	NA
	Density less than minimum specification (less than 2.00 points per square unit)	40.00
	Density within specification (2.00 to 10.00 points per square unit)	60.00
	Density exceeds specification (more than 10.00 points per square unit)	0.00

First Return (Pulse) Density

Density computed using a cell that is 316.23w by 316.23h units (100000.00 square units)

First return density image is 235 rows by 14 columns: test_batch_first_return_density.jpg

Minimum first return density (returns per square unit)	0.00
Maximum first return density (returns per square unit)	5.93
Average first return density (returns per square unit)	2.23
Standard deviation of first return density (returns per square unit)	0.92

Color	Description	Percentage of Area with Data
	Cells with no points (could be outside coverage area)	NA
	Density less than minimum specification (less than 2.00 points per square unit)	42.19
	Density within specification (2.00 to 10.00 points per square unit)	57.81
	Density exceeds specification (more than 10.00 points per square unit)	0.00

Figure 14: Point density test results using point in cell method but with varying cell sizes (highlighted). Results are binary pass/fail statistics (courtesy Matt Bethel).

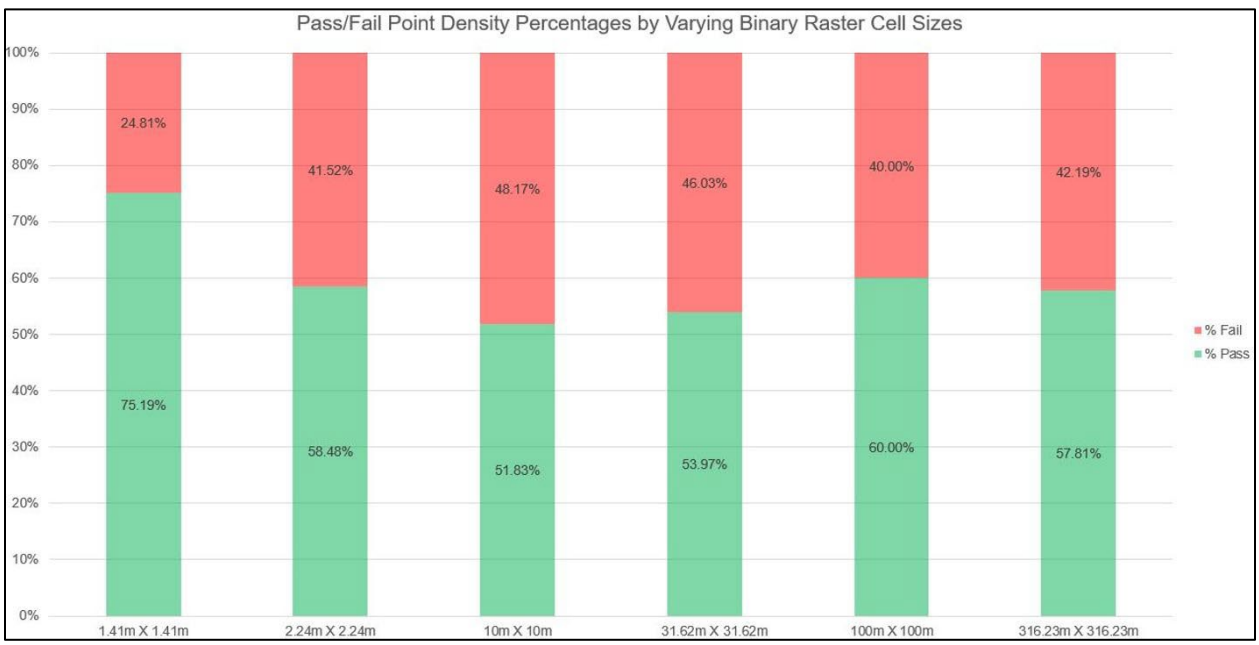

Figure 15: The results from Figure 14 charted as pass/fail percentages (courtesy Matt Bethel).

- Advantages
  - Straightforward approach
  - Fast and easy to calculate
  - Easy to use for reporting – pass/fail percentage results and graphics
  - Performs consistently with all lidar scanner types and all acquisition approaches (e.g., >50% overlap or multiple channels/sensors)

- Disadvantages
  - The results are in pass/fail cell counts yet there are no established parameters for use or analysis (no passing thresholds)
  - Different user-defined processing cell sizes significantly change the results
  - Inherent with aliasing problems that invalidate the results
  - These disadvantages are most prominent at fringe density cases which coincide with the most cost-efficient way to collect lidar data.

5.7 Voronoi – A representative area for each point is constructed using Voronoi/Thiessen polygon methodology. Density is derived by calculating the inverse of the area. Statistics of these densities are reported as well as a thematic raster and pass/fail areas.

- Methodology
  1. Create Voronoi polygons, one for each last return point or pre-conditioned point for measurement if no return values exist.
  2. Calculate areas per polygon.
  3. Calculate the inverse of the area to attribute each polygon with a calculated density.

The following diagrams are steps 1-3 from the methodology listed above:

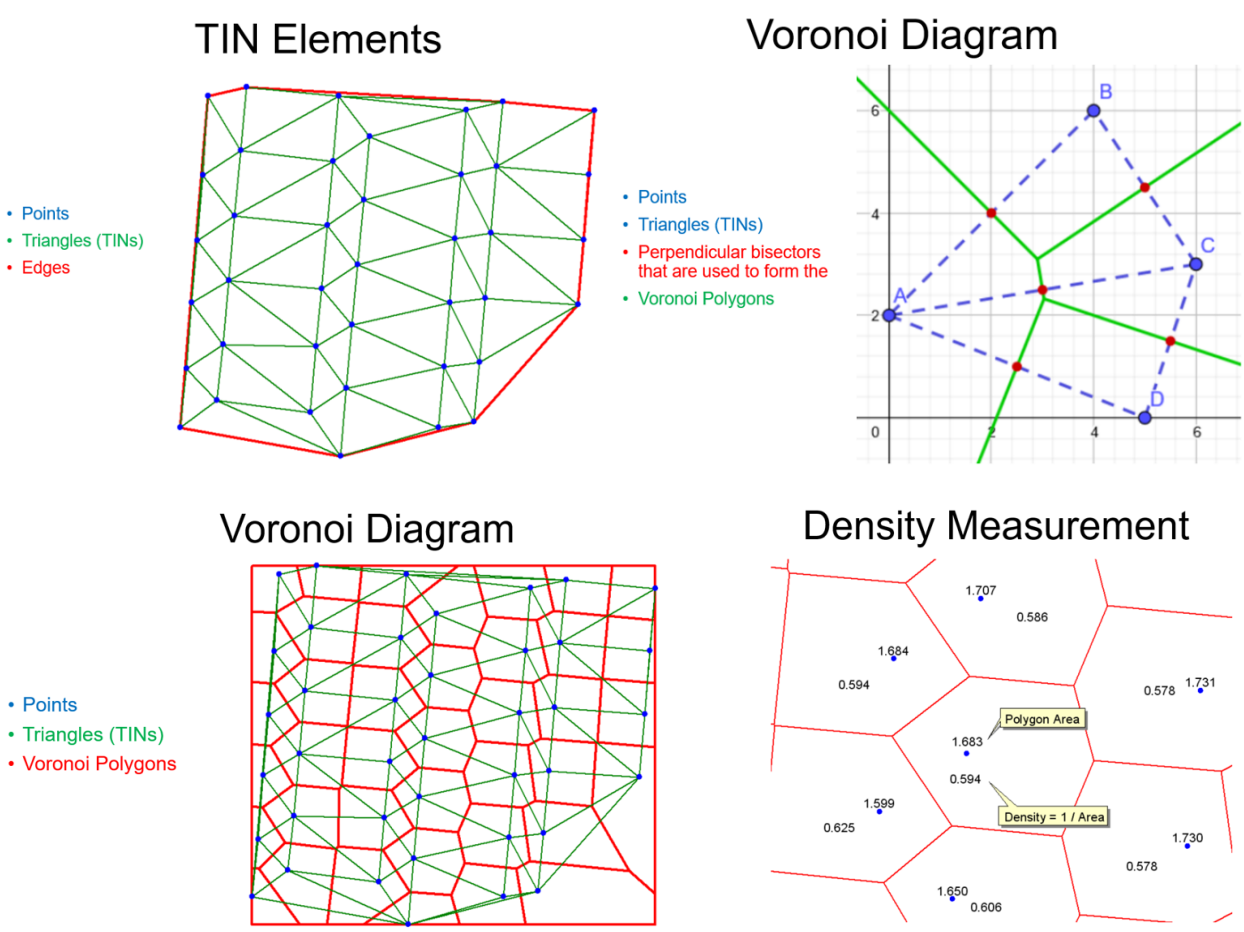

Figure 16: Series of four diagrams showing the construction of the Voronoi model leading to point density measurements (courtesy Ty Naus: "Nominal Point Spacing and Accuracy Assessments" 2011).

4. Bin all values from step 3 using the Freedman–Diaconis rule to create a frequency distribution chart of point densities (Figure 17). High density cutoff is 4x design density by default.
5. Rasterize the polygons at 0.5 x NPS (NPS is calculated from the user input of design density). The raster values are generated using polygon to raster processing with the density values calculated from step 3 as input.
6. Thematically color this raster to show density variations across the tested dataset (Figure 18).
7. Create a binary pass/fail raster that is 100 x NPS (Figure 19). This larger cell size is used to aggregate densities into meaningful geographic bins of modal measurements so as to minimize the affect of localized lidar shadowing from buildings, trees, and other above ground features.
   - Code each raster cell with the mode of the pixel values of the extent of the cells from step 5, setting cells to No Data that meet the following qualifications:
     - No Data values from water body clusters (user-definable) and/or exclusion polygons
     - Binary raster cells that are not completely covered in raster cells from step 5
   - Thematically color this binary raster with green for passing cells (mode is ≥ user inputted design density) and red for failing cells (mode is < user inputted design density).
8. Summarize the percent pass/fail tiles. Greater than or equal to 90% is an ideal guideline for an acceptable tested block of point cloud data.

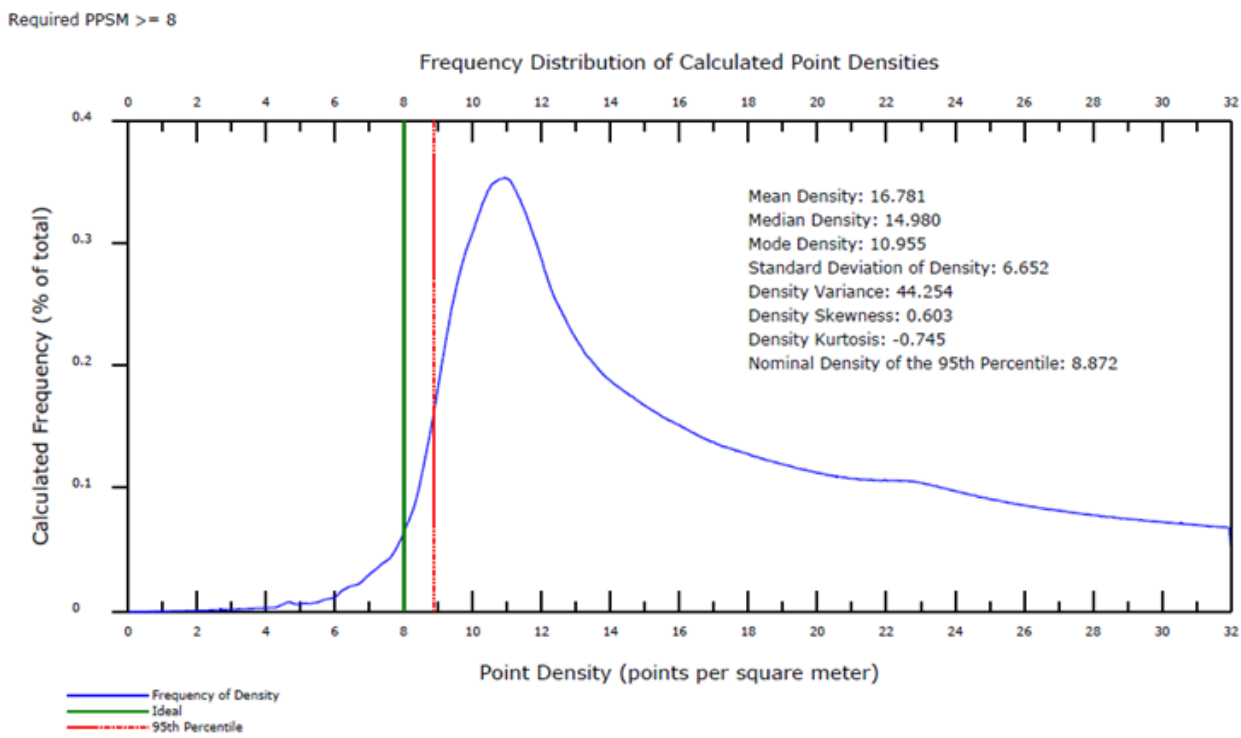

Figure 17: Voronoi density frequency distribution chart (courtesy Matt Bethel).

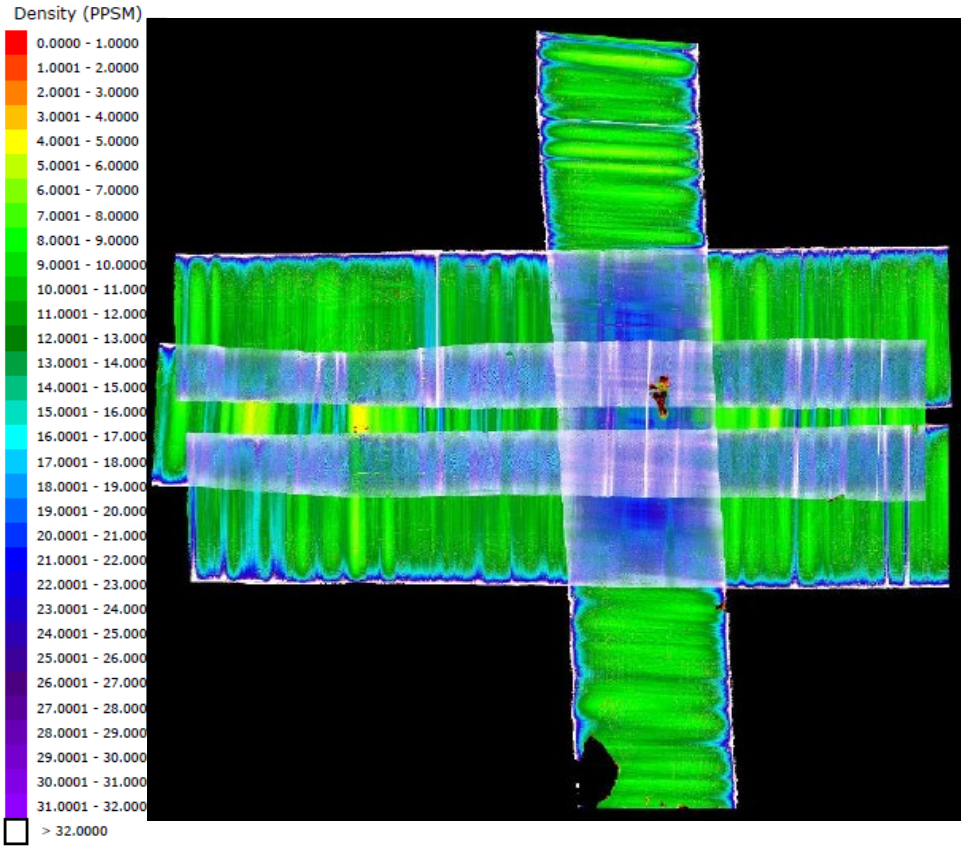

Figure 18: Voronoi density thematic raster (courtesy Matt Bethel).

Figure 19: Voronoi density pass/fail raster (courtesy Matt Bethel).

Since the Voronoi method does not suffer from the problems of the point in cell method, varying cell sizes to measure the representative density has little affect on the pass/fail percentages as shown below:

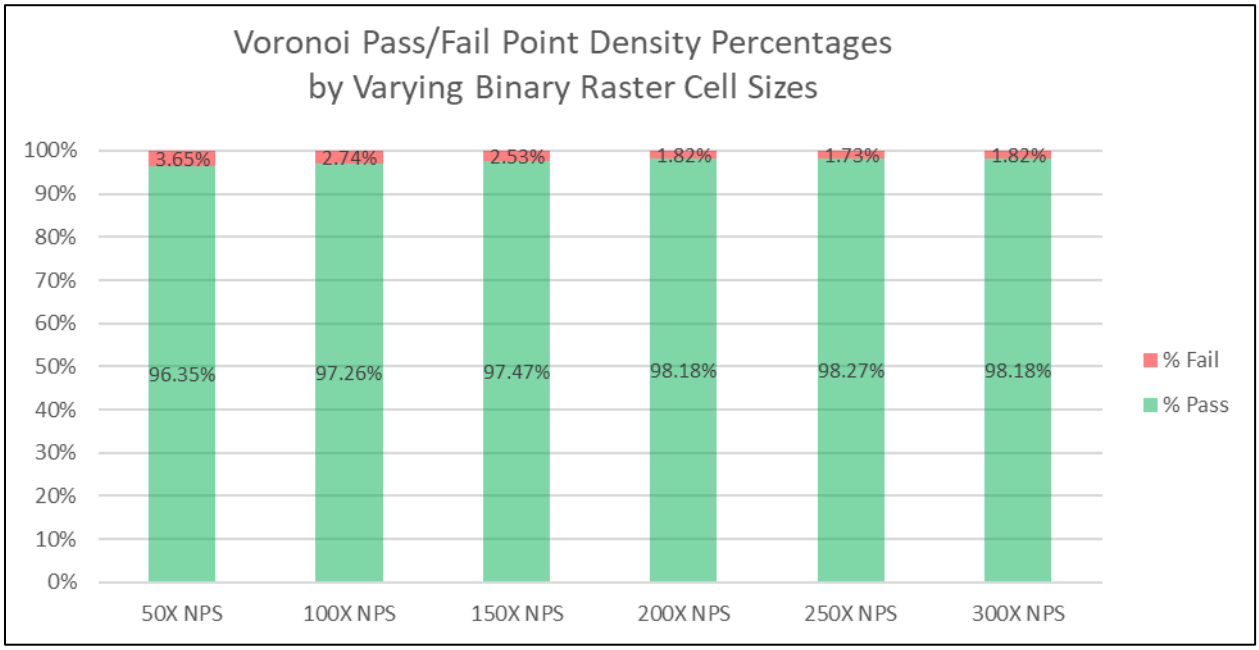

Figure 20: 8 ppsm pass/fail percentages charted for Voronoi tests using different cell sizes (courtesy Matt Bethel).

- Advantages
  - Most accurate representation of point density
  - Measurement is an area of point influence. Density can be derived by 1/Voronoi area.
  - Performs consistently with all lidar scanner types and all acquisition approaches (e.g., >50% overlap or multiple channels/sensors)
  - Is not affected by aliasing or varying tile sizes
  - Pass/fail is not strongly biased by isolated cell failures like building shadows
  - Pass/fail is not biased by scanner type, overlap, crosslines, or patches
  - Preserves decimal precision rather than being limited to integer values
- Disadvantages
  - Longer processing time than previous methods but can be parallelized and/or distributed to mitigate.
  - Limited software support

5.8 Pre-Conditioned Planar Surface Density Assessment – This method pre-conditions point clouds lacking return information such as Geiger Mode Lidar (GML) to avoid inflating density estimation due to isotropic clutter and sampling multiple surfaces in the vertical direction. Expected inter-point spacing is computed for all points in three-dimensional space. The computed inter-point spacing is used to establish and select only locally planar surfaces in three-dimensional space to participate in the density assessment. Expected three-dimensional nearest-neighbor distance is computed for selected points. Points are projected to the horizontal plane and duplicates are eliminated based on a stratified sampling at the expected nearest-neighbor distance. Density estimates are computed as in the Voronoi approach. Assessed point cloud density can be based on either mode or cumulative density depending on whether expected or threshold performance is desired.

- Methodology
  1. Compute mean all-points nearest-neighbor spacing (s).
  2. Compute local dimension for each point based on a spherical radius of $2\sqrt{2}s$. Local dimension is computed as the dimension associated with the largest jump in singular value resulting from a principal components analysis of the local neighborhood with singular values associated with dimension in decreasing order.
  3. Down-select point cloud to points with dimension 2.
  4. Re-compute mean all-points nearest-neighbor spacing for selected points ($\hat{s}$).
  5. Project selected points to the horizontal plane.
  6. Perform a two-dimensional face-centered cubic (regular hexagonal raster) stratified sampling with raster spacing of $\hat{s}$ to remove residual duplicate points.
  7. Create Voronoi tessellation of conditioned point cloud.
  8. Cull all Voronoi polygons that intersect the convex hull of the projected point clouds and remove associated points from subsequent analysis.
  9. Calculate area of Voronoi polygon associated with each point.
  10. Compute local density for each point as inverse of associated Voronoi polygon area.
  11. Assessed density can be established either via kernel density estimation, peak bin from histogram with bin width established according to the Freedman-Diaconis rule, or CDF intersection from density population depending on desired assessment.
  12. (Optional) Thematic maps can be generated based on locally assessed density values for qualitative assessment.

Advantages
- Identical to Voronoi approach
- Removes influence of isotropic clutter and multiple surface sampling from assessment of horizontal density

Disadvantages
- Requires the most complex conditioning workflow, but can be greatly sped up with appropriate use of metric space data structures and parallelization
- Longer processing time than previous methods but can be parallelized and/or distributed to mitigate.
- Limited software support

6. Methodology Comparison

	Sensor and scanner type agnostic	Results are meaningful with respect to the entire project	Can be used to easily identify valid, reliable, localized problems anywhere in the project	Not flawed by raster cell aliasing	Not flawed by varying tile sizes for summary reporting	Pass/fail results are not negatively biased (inflated) by overlap, crosslines, patch lines, etc.	Easily used if flying dual channel, dual sensor, or >50% overlap configurations to meet the minimum design density	Project-wide measurement and reporting	Is scalable, batchable, and distributable
Representative Sample	Y	N	N	Y	Y	N	Y	N	N
Per Swath	N	N	N	Y	Y	Y	N	Y	Y
Aggregate	Y	N	N	Y	Y	N	Y	Y	Y
Point In Cell Raster Count	Y	N	N	N	N	Y	Y	Y	Y
Compute Statistics for the Sweet Spot of the Swath	N	N	N	N	N	Y	N	Y	Y
Binary Raster	Y	N	N	N	N	Y	Y	Y	Y
Voronoi	Y	Y	Y	Y	Y	Y	Y	Y	Y
Pre-Conditioned Planar Surface Density Assessment	Y	Y	Y	Y	Y	Y	Y	Y	Y

7. Existing and Common Density Use Cases

- Post collection check and confirmation for required conformity
- To produce a contiguous product (e.g., grid) that is appropriate for the measured horizontal sampling density
- To determine what derivative products can be produced from a point cloud dataset
- To determine the finest detail to use as input into algorithm settings or parameters
- Measuring density under vegetation using ground classified points
- Comparison of temporally different point cloud datasets
  - To determine the common denominators (such as cell size) to use for comparison/analysis
  - To help decide which dataset to use when two or more collects cover the same area
- Comparison of different sensor modalities of remotely sensed data

8. Recommendations

8.1 Design Density

Since density of unstructured and unorganized point cloud varies, the metric must be considered a distribution, not a single number. If a lidar acquisition is planned for the minimum required density, it is likely that a significant portion of the acquired data will fail an appropriate density test. In the example below if a user plans for an average design density of 8 ppsm and the project requirement is 8 ppsm (meaning a required minimum of 8 ppsm), approximately half the data could be <8 ppsm (red) and the other half >8 ppsm (green) as depicted in Figure 21. If the flight planning takes this into consideration and over-plans the average design density (blue line), the graphic depicted in Figure 22 is more likely.

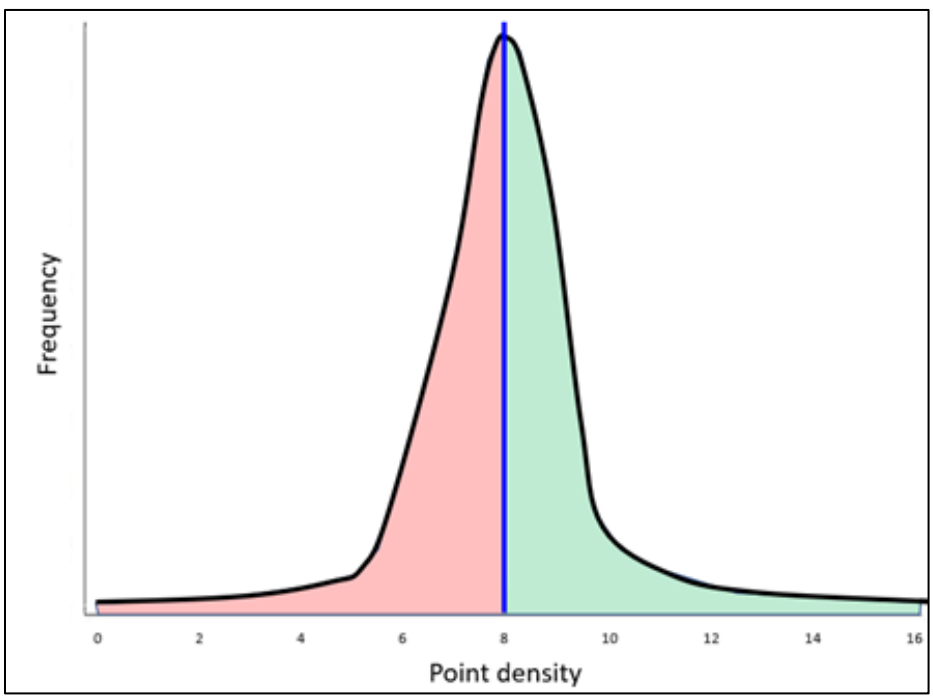

Figure 21: Frequency distribution chart of a targeted point density average equaling 8 ppsm (courtesy Matt Bethel).

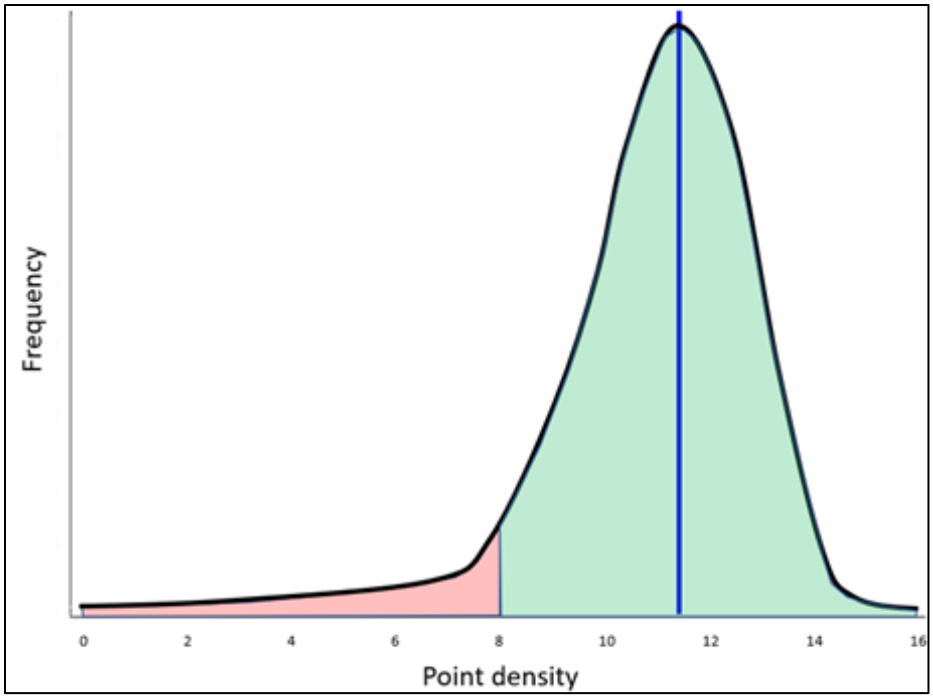

Figure 22: Frequency distribution chart of a targeted point density for most of the data to be greater than 8 ppsm (courtesy Matt Bethel).

8.2 Testing Density

The recommended method for measuring and reporting horizontal sampling density of aerial lidar point cloud data is the **Voronoi method** for linear mode lidar and the **Pre-Conditioned Planar Surface Density Assessment method** for point clouds that do not have return information such as Geiger Mode Lidar (GML). Both recommendations use the most accurate density measuring approaches possible. Neither suffers from the limitations of all the other methods reviewed in this paper. And both provide a report that includes 1) a frequency distribution chart, 2) robust statistical information about the density distribution, 3) a thematic raster, and 4) an effective pass/fail binary raster assessment that is focused on clusters of failing modal values while not being heavily influenced by failing cells that naturally occur in most lidar datasets.

However, the Voronoi method is computationally more intensive than the other methods discussed in this document (O(nlog(n)) vs O(kn) where k is the number of neighbors). This implies that as the density of points increases, the performance of the Voronoi method in terms of speed decreases. To mitigate this issue, an approach such as segmenting data into virtual tiles while processing should be used so that each virtual tile has similar performance, and the speed is O(t) where t is the number of virtual tiles.

Other options that can be considered is to measure the density from a sub sampled number of points, parallelization of the processing as well as faster implementation methods which are available in the literature. For e.g., Voronoi method generates vertices that do not exist in the point cloud. The Delaunay triangulation can therefore be used to estimate the area of Voronoi triangles.

From a product perspective, the Voronoi (and its dual, the Delaunay triangulation) are useful higher-level products. Many users require the delivery of these products, along with the point cloud; therefore, the added effort to generate density via this recommended method can be considered well spent.

It is recommended that software vendors that provide lidar data quality assessment algorithms provide the users with an option to assess lidar data with the Voronoi based methods of density assessment.